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Latitudinal patterns and environmental drivers of moss layer cover in extratropical forests

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Abstract

Aim: The forest moss layer plays an important role in water retention, nutrient cycling and forest regeneration. We quantified broad-scale latitudinal patterns and environmental drivers of moss layer cover in extratropical upland forests where such information has been missing.

Location: One thousand, nine hundred and seventy-seven sites outside the tropical belt (i.e., beyond 26° N and S) across longitudes (from 154° W to 140° E) and elevations (from 0 to 3,950 m a.s.l.).

Time period: Current (1969–2016).

Major taxa studied: Moss layer bryophytes (mosses, liverworts and hornworts).

Methods: We compiled three large datasets from: (a) a global literature search, (b) a national forest inventory database in the USA and (c) an extensive survey along the Appalachian Trail (Eastern USA) to contrast the moss layer cover along latitudinal gradients and among regions. We used correlation analysis and simple (meta- and linear-) regression to test whether moss layer cover increased towards the poles across and within forest biomes, and we used multiple (meta- and beta-) regression to test whether and how climate, forest composition and age and topography influence forest moss layer cover at broad spatial scales.

Results: Forest moss cover increased towards the North along a clear latitudinal gradient by c. 2% per 1° of latitude, and it decreased with increasing maximum temperature by c. 1%–5% per 1°C. Thus, cool northern regions (e.g., northern Europe) can be considered forest moss layer hotspots. Moss layer cover was driven mainly by canopy composition when all potential drivers were considered. Compared with broad-scale systematic surveys, the data from specialized literature overestimated moss cover in broadleaf deciduous forests where moss cover was low and variable.

Main conclusions: Our findings highlight that systematic surveys complement specialized literature in accurately modelling abundance on broad scales. Given that temperature, rather than precipitation or seasonality, limited the abundance of forest moss layer on broad scales, the moss layer might be a promising tool for monitoring forest ecosystem responses to climate warming.

KEYWORDS

beta regression, bryophytes, Forest Inventory and Analysis (FIA), latitudinal gradients, meta-regression, temperature, WorldClim

1 | INTRODUCTION

Bryophytes, a group of non-vascular land plants that include mosses, liverworts and hornworts, are broadly distributed among terrestrial ecosystems (e.g., Hedenäs, 2007; Medina, Draper, & Lara, 2011), where they play important functional roles (Cornelissen, Lang, Soudzilovskaia, & During, 2007; Lindo & Gonzalez, 2010) while being increasingly threatened by global warming (He, He, & Hyvönen, 2016). In extratropical forests (i.e., mid- and high-latitude forests; see Elbert et al., 2012), bryophytes are more abundant at the ground level—where they form a moss layer—than in the other vertical strata (Elbert et al., 2012). The moss layer plays important roles in influencing soil physical properties (e.g., Soudzilovskaia, Van Bodegom, & Cornelissen, 2013), forest regeneration (Soudzilovskaia et al., 2011), ecosystem water budgets (Bond-Lamberty, Gower, Amiro, & Ewers, 2011) and mineral cycles (e.g., Gundale, Deluca, & Nordin, 2011; Nasr & Arp, 2011). The sensitivity of the moss layer to ongoing climate warming (He et al., 2016) highlights the importance of understanding both the distribution of the moss layer on broad spatial scales and the effects of the main climatic and other environmental drivers on moss layer distribution at these broad scales (cf. Chen et al., 2015; Suding et al., 2008; Zhang, Chen, & Taylor, 2016).

Compared with the distribution of vascular plants, the distribution of the moss layer has been studied less frequently at broad spatial scales. For example, the moss community was found to increase in species richness across large latitudinal gradients towards the poles in both hemispheres; this pattern was found across 25° of latitude in Chile (Rozzi et al., 2008) and 35° of latitude in Western Europe (Mateo et al., 2016), and a similar but only marginally significant pattern was found across 30° of latitude in China (Chen et al., 2015). Compared with moss species richness, overall moss layer abundance (usually measured as the percentage of ground covered by the moss layer; hereafter, moss cover) has been studied at broad spatial scales less frequently, although it is less time consuming and less costly to quantify and is frequently used in major national monitoring programmes (e.g., U.S. FIA Program; O'Connell et al., 2015). Thus, published studies of the moss layer seem biased towards alpha diversity rather than changes in the abundance or distributions at broad scales (cf. McGill, Dornelas, Gotelli, & Magurran, 2015). Monitoring moss cover is particularly important because it affects ecosystem functions such as water retention (Bond-Lamberty et al., 2011) and nutrient cycling (Elbert et al., 2012).

Forest mosses are well adapted to marginal oligotrophic environments (Bates, 2000), but they are physiologically limited by water availability (Proctor et al., 2007) and sensitive to forest disturbance (e.g., Dovčiak, Halpern, Saracco, Evans, & Liguori, 2006). At fine (microsite) scales, forest moss cover tends to increase with moisture (e.g., Busby, Bliss, & Hamilton, 1978), complexity of the microtopography (Økland, Rydgren, Økland, Storaunet, & Rolstad, 2003) and time since disturbance (forest age; e.g., Liu & Bao, 2014; Zhang et al., 2016). In addition, moss cover tends to decrease with nutrient and light availability (Glime, 2007), and it varies with substrate type (Halpern, Dovciak, Urgenson, Evans, 2014) and canopy composition

(Bartels & Chen, 2013; Hart & Chen, 2006). As the moss layer has traditionally been studied at fine spatial scales (measured in square metres or square centimetres; e.g., Botting, Campbell, & Fredeen, 2008; Lencinas, Pastur, Solán, Gallo, & Cellini, 2008), much less is known about the ecological drivers of moss cover at broad spatial scales described by latitudinal or elevational gradients in climatic variables such as temperature and moisture (Chen et al., 2015; He et al., 2016). Consequently, our understanding of how changes in broad-scale climate or other ecological factors may affect the global and regional distributions of forest moss layer is limited (cf. Zhang et al., 2016).

We studied forest moss cover in extratropical upland forests to improve the understanding of the latitudinal patterns and ecological drivers of forest moss abundance. We analysed three datasets collected at three broad spatial scales: (a) intercontinental scale (extratropical regions of the world; ETR dataset), (b) continental scale (USA; US dataset) and (c) subcontinental scale (the Appalachian Trail in the Eastern USA; AT dataset). Our objectives were as follows: (a) to contrast moss abundance in upland forests across extratropical world regions and biomes and (b) to explore how latitudinal climatic gradients affect forest moss cover at these different broad scales. Hence, we tested three hypotheses (hypotheses 1–3) about moss cover and its environmental drivers; we hypothesized that forest moss cover increases towards the poles across (hypothesis 1) and within (hypothesis 2) extratropical biomes (consistent with the patterns in moss richness), and that temperature and precipitation are the main drivers of moss cover at broad spatial scales (across latitudes) rather than finer scale site factors such as forest composition, topography or time since stand-replacing disturbance (forest age) (hypothesis 3). To our knowledge, this is the first effort to summarize quantitatively the spatial distribution of forest moss cover (as opposed to richness) relative to major environmental drivers in extratropical biomes across latitudinal gradients on global to regional spatial scales (but for global patterns of cryptogam cover, see Elbert et al., 2012). We focus on upland forests where moisture and temperature limitations are likely to play a larger role; we exclude *Sphagnum*-dominated lowland forests and peatlands where hydrology has unique characteristics (modified by *Sphagnum* communities; Bragg, 2002) and where the role of climate has been studied more widely (e.g., Limpens et al., 2011).

2 | METHODS

2.1 | Study design

We assembled three datasets (Supporting Information Appendix S1) by compiling data on forest moss cover and forest stand structure (composition, age) collected at three broad spatial scales (intercontinental, continental, and subcontinental; Table 1 and Figure 1) and linking them to climate, biome, and region using site geographical coordinates. Thus, each dataset contained the observed forest moss cover (the response variable) and its potential environmental drivers

TABLE 1 Comparison of the three datasets assembled to examine moss cover distribution relative to major environmental factors in extratropical forests

Dataset (scale)	Sample area (m ²)	Extent (m ²)	Number of plots	Data source
ETR (intercontinental)	1 (0.01–100) ^a	360 (16 to 296 × 10 ⁴) ^b	82	Literature ^c
US (continental)	168	4,000	1,865	FIA ^d
AT (subcontinental)	1	3,600	30	Field survey ^e

Notes. AT = dataset assembled from a region-wide systematic survey; ETR = dataset assembled using a literature search; Extent = area used to average moss cover across individual plots at the forest site level; Sample area = area used to estimate moss cover; US = dataset assembled using the systematic FIA survey in the USA. Note. Plot distribution is shown in Figure 1.

^aMode and range of the plot size in the ETR dataset.

^bMedian and range of the extent in the ETR dataset.

^cDetails are in Supporting Information Appendix S2.

^dForest Inventory and Analysis (FIA) National Program of the U.S. Forest Service.

^eDetails are given by Lawrence et al. (2015).

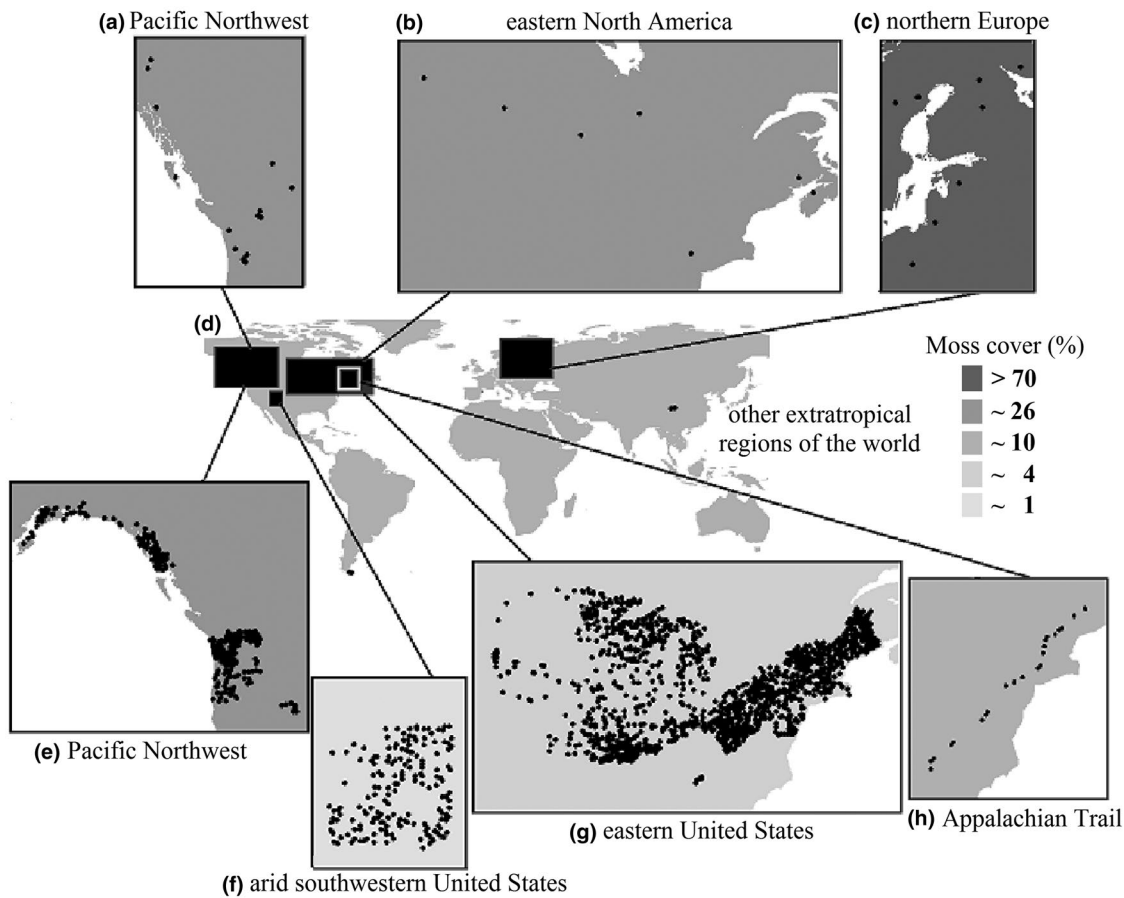


FIGURE 1 Plots with forest moss cover (as a percentage) reported in the literature for extratropical regions of the world (ETR, intercontinental scale dataset; a–d; $n = 82$; sources available in Supporting Information Appendix S2) or in a nationwide systematic survey (US, continental scale dataset; e–g; $n = 1,865$ FIA Phase 3 plots; FIA = the Forest Inventory and Analysis National Program of the USA; note that FIA Phase 3 data were not available for all U.S. states) or in a systematic survey along the Appalachian Trail (AT, subcontinental scale dataset; h; $n = 30$ Appalachian Trail Mega-Transsect plots). Note that (d) represents the baseline map depicting the regional insets in addition to several individual plots identified as a part of the ETR dataset outside the main regions. Dots in panels (a–h) represent forest plots (including a few isolated plots in panel d found in the literature in Eastern Asia and South America). Black boxes in panel (d) show data clusters grouped into world regions. Mean moss cover in each region is represented by shading intensity. Panels are projected from plot coordinates (available in Supporting Information Appendix S1) as Lambert azimuthal equal area

(Table 2) as well as site spatial attributes (geographical coordinates, biome and region).

2.1.1 | Datasets

The ETR dataset was assembled from published literature using a systematic search with predefined criteria to ensure procedural objectivity and transparency (Haddaway, Woodcock, Macura, & Collins, 2015), and the data were integrated and analysed using a meta-analysis approach (Borenstein, Hedges, Higgins, & Rothstein, 2010). We searched the ISI Web of Science database (Reuters, 2013) using a systematic query '[(bryophytes OR mosses) AND ecology AND forests AND understory] AND [abundance OR biomass OR cover]' in the 'topic' field and retrieved 245 papers (published between 1969 and 2016). From this set of papers, we selected those that reported forest moss cover in extratropical regions (too few studies reported forest moss biomass) and excluded those that (a) included lichens in the moss cover, (b) were devoted to only one species, (c) focused on *Sphagnum*-dominated ecosystems (peatlands) or (d) were laboratory studies. We further limited the set of selected papers to those that reported moss cover mean, its variability and sample size—the variables required for meta-analyses in order to integrate data collected by different authors using different methods (Borenstein et al., 2010). If data were reported graphically, we processed the figures with Engauge Digitizer 4.1 (Mitchell, 2002) to obtain numerical values. The final ETR dataset for analyses consisted of 82 forest plots gathered from 30 papers published between 1995 and 2016 (the list of the data sources is in Supporting Information Appendix S2). We grouped data into four world regions based on their location (Pacific Northwest, eastern North America, northern Europe and other extratropical regions; Figure 1a–d) and into biomes based on Olson et al. (2001). Three biomes were present in the ETR dataset: (a) boreal forest and taiga (hereafter Boreal), (b) temperate conifer forest (hereafter Conifer) and (c) temperate broadleaf and mixed forest (hereafter Broadleaf) (Supporting Information Appendix S2).

The US dataset was assembled from the data collected by the Forest Inventory and Analysis (FIA) National Program of the United States Forest Service (O'Connell et al., 2015), available online (Forest Inventory & Analysis, 2014). This programme surveys forest plots systematically placed on forest land across the United States to inventory forest stand characteristics, individual tree measurements, and vegetation indicators. The FIA includes three phases of sampling using 0.4 ha plots, each composed of four 168 m² circular subplots. Only FIA Phase 3 includes moss cover data, and the data for this phase are not available for all USA states. Our dataset contained data from 1,865 FIA plots spread across the continent, which we grouped based on their location into three US regions: Pacific Northwest, eastern United States and arid Southwestern United States (Figure 1e–g). Based on Olson et al. (2001), these locations were distributed in five biomes; three forest biomes and two non-forest biomes containing azonal forest vegetation: (a) Boreal, (b) Conifer (temperate), (c) Broadleaf (temperate), (d) Desert and xeric shrublands (hereafter Azonal forests in deserts) and (e) Temperate grasslands, savannas and shrublands (hereafter, Azonal forests in grasslands).

The AT dataset was assembled from a field study of forest understorey plant communities along the Appalachian Trail, which is one of the longest hiking trails in the United States (c. 3,500 km), leading along the crest of the Appalachian Mountains and crossing 14 states in the eastern United States (the AT Mega-Transect Study; Lawrence et al., 2015). The dataset contains 30 mesic forest plots randomly located in sugar maple, oak, and spruce-fir forests within 3 km of the Appalachian Trail, with the southernmost plot located in Coweeta Long Term Ecological Research (LTER) site in North Carolina and the northernmost plot in the Appalachian Mountain Club 100 Mile Wilderness Conservation Area in Maine (Figure 1h). The AT plots are in the eastern United States and, based on their canopy composition, they correspond to two biomes: (a) Conifer, and (b) Broadleaf.

Environmental driver	Variable tested
Forest age	Forest age in years (time since stand-replacing disturbance)
Canopy composition	Forest type: conifer, mixed ^a , hardwood
Climate	Mean annual temperature, summer temperature, winter temperature, maximum temperature, minimum temperature, annual precipitation, precipitation in the driest month ^b
Topography	Elevation (m a.s.l.) ^c

^aMixed conifer and deciduous hardwood species without a clear dominance of either of the two types (cf. Supporting Information Appendix S2 and the FIA descriptions available online: <https://www.fia.fs.fed.us/library/database-documentation/>).

^bClimatic variables obtained from WorldClim (Hijmans et al., 2005; www.worldclim.org/bioclim) represent annual means or totals (mean annual temperature, annual precipitation), seasonality (summer temperature: mean temperature of the warmest three-month period; winter temperature: mean temperature of the coldest three-month period) and limiting environmental factors (maximum and minimum temperature; precipitation of the driest month).

^cObtained from field surveys in the US and AT datasets and from the selected papers or global climate model (WorldClim; Hijmans et al., 2005) in the ETR dataset.

TABLE 2 Environmental drivers of moss cover and specific variables tested

2.1.2 | Measurement of forest moss cover

Most of the literature sources (94%) compiled into the ETR dataset estimated moss cover visually as a percentage of plot area covered with moss; the point intercept method was used by 6% of the sources. The most common sample unit size was 1 m² (48%), followed by 0.01 m² (14%) and 0.02 m² (10%). The total area used to average moss cover at the forest stand level in each study varied considerably, from 16 to 296 × 10⁴ m² (Table 1; Supporting Information Appendix S2).

In the US dataset, we summarized moss cover by calculating mean values for those 0.4 ha FIA plots with data for at least two 168 m² subplots. FIA Phase 3 crews visually estimate total moss cover on one to four subplots, depending on terrain suitability (details in Forest Inventory & Analysis, 2014). We selected FIA Phase 3 plots surveyed recently (i.e., between 2000 and 2011) where the subplots belonged to the same forest type in order to exclude plots placed on boundaries between vegetation types. We also excluded hydric plots (see Forest Inventory & Analysis, 2012) as a conservative measure to exclude forests with moss layer dominated by *Sphagnum* species (as we did in the ETR dataset). Most FIA plots included in the US dataset contained four subplots (83% of the dataset), and a few contained three (11%) or two (6%) subplots.

Forest moss cover in the AT dataset was estimated at each site as the average percentage of ground covered by mosses in 27 quadrats arranged in nine clusters (plots) located on a grid with 25 m spacing between the clusters; each cluster consisted of three 1 m² understorey quadrats placed 2 m apart (additional details are provided by Lawrence et al., 2015).

2.1.3 | Environmental drivers

We gathered geographical coordinates, data on canopy composition, and topography (Table 2) for each forest plot included in the ETR dataset directly from the selected papers; when elevation was not reported in the original source, it was obtained from the global climate model (WorldClim; Hijmans, Cameron, Parra, Jones, & Jarvis, 2005) by using the latitude and longitude of the forest plot. Likewise, in our US dataset, we collected plot geographical coordinates and summarized data on stand characteristics and topography from the FIA database by calculating plot means for quantitative data and modes for qualitative and nominal data. Finally, for the plots included in the AT dataset, we recorded their geographical coordinates and their canopy composition (conifer, hardwood) and elevation (topography). Although all plots were in mature forests, we did not relate forest age to moss cover owing to relatively low replication of young and old ages across the forest types in the AT dataset. By using the geographical coordinates of each forest plot, we acquired climatic data (Table 2) from the global climate layers modelled by WorldClim 1.4 with resolution of c. 1 km² (Hijmans et al., 2005).

2.2 | Statistical analysis

2.2.1 | Synthesizing data to contrast moss cover across regions

The three datasets described above varied in their character, and thus they required different statistical tools and data processing to contrast the moss cover among the regions (objective 1). We applied a random-effects model (REM) meta-analysis to synthesize the information from the ETR dataset following Haddaway et al. (2015) by using the reported mean of moss cover transformed to ratio (cover/100) as the effect size. In REM meta-analysis, the study is a random effect because it is assumed that the response varies among different studies, and datasets from studies with large and small *n* are made comparable by weighting each dataset with its variance (i.e., low-variance data are weighted more than high-variance data; Borenstein et al., 2010). Moss cover variance (reported or calculated from reported *n* and *SD*, *SE* or range) was also transformed to a ratio to scale it properly. Given that the test for heterogeneity (Borenstein, Hedges, Higgins, & Rothstein, 2009) rejected homogeneity (*Q sensu* Borenstein et al., (2009) is the statistic to test homogeneity among studies; it is sensitive to the ratio of observed variation to the within-study error: *Q*₈₁=21,161; *p* > 0.001), we calculated the pooled mean of moss cover and its corresponding *SE* by world regions (Figure 1a–d) and tested the difference in moss cover between the regions for significance using Wilcoxon rank sum tests with Bonferroni correction for multiple testing ('stat' package; R Core Team, 2013). As our literature search found only six sites that fitted our search criteria in the 'other' extratropical regions (Figure 1d), we excluded these sites from further analyses owing to this low sample size.

To synthesize the information from the US dataset, we calculated the mean and *SE* of moss cover in different regions of the United States (Figure 1e–g) and tested for differences among them with Kruskal–Wallis tests because moss cover was not normally distributed (Shapiro–Wilk normality test: *W* = 0.47, *p* < 0.001, *n* = 1,865); when the Kruskal–Wallis test identified significant differences (*p* < 0.05), we identified statistically different means with a Wilcoxon rank sum tests with Bonferroni correction ('stat' package; R Core Team, 2013). Finally, we summarized the information collected at the regional scale along the Appalachian Trail (Figure 1h) by calculating the mean and *SE* of moss cover for the two biomes (Broadleaf and Conifer) represented along the steep elevational climatic gradients and we tested for differences between them with the Kruskal–Wallis test because moss cover was not normally distributed (Shapiro–Wilk normality test: *W* = 0.79, *p* < 0.001, *n* = 30). To compare moss cover means estimated for a particular region from the systematic surveys (US and AT datasets) statistically, we applied a *t*-test with unequal variances. The moss cover estimates from the ETR dataset were compared with the estimates from the other two datasets visually because the meta-analysis nature of the ETR dataset precluded a direct statistical comparison.

2.2.2 | Hypothesis testing

We used correlation and regression analyses to test the effects of latitude on forest moss cover across and within biomes (objective 2, hypotheses 1 and 2). To test whether moss cover increased polewards across extratropical biomes (hypothesis 1), we performed these analyses on the datasets containing more than two biomes: the ETR and US datasets. When we found a statistically significant correlation ($p < 0.05$), we fitted one of two types of regression models to quantify the rate of change in forest moss cover per latitudinal degree; we used (a) a random effects meta-regression ('metafor' package; Viechtbauer, 2010) with the ETR dataset collected from the literature and (b) a simple linear regression ('stat' package; R Core Team, 2013) with the US dataset based on FIA field data. To test whether moss cover increases polewards (towards the North Pole, given that all of the data included in the analyses were from the Northern Hemisphere) within individual biomes (hypothesis 2), we performed the correlation and regression analyses as described above on all three datasets (ETR, US and AT) by biome (based on Olson et al., 2001; see above). We evaluated the spatial dependence of moss cover within each biome by calculating the Moran's I statistics with the 'spdep' R package (Bivand et al., 2006).

To test whether climate is the main driver of moss cover across latitudes at broad spatial scales (hypothesis 3), we first identified climate variables consistently correlated with forest moss cover. A correlation analysis between forest moss cover and seven climate variables (Table 2) was carried out on all three datasets (ETR, US, and AT) using linear (Pearson) and nonlinear (Spearman) correlations (ρ) calculated with the R package 'stat' (R Core Team, 2013). To choose a subset of the most relevant climatic variables consistently related to moss cover, we selected all climatic variables with statistically significant ($p < 0.05$) and consistent (i.e., positive or negative) correlations with forest moss cover in all three datasets (ETR, US, and AT), where at least one of the datasets showed at least medium correlation ($|\rho| > 0.35$; cf. Cohen, 1992) with moss cover. We applied these correlation thresholds ($|\rho| > 0.35$, $p < 0.05$) also for the corresponding variables of canopy composition, forest age and topography (Table 2) to reduce predictor sets to the most relevant variables in each dataset (ETR, US, and AT). Finally, we used multiple regression to identify the most parsimonious models as explained below.

For the ETR dataset, we fitted REM meta-regression models similar to generalized linear models (GLMs) by including fixed effects that extended the REM meta-analysis by integrating data collected by different authors (Borenstein et al., 2009). The full model was defined as.

$$f(y) = \beta_0 + \beta_{\text{Climate}} + \beta_{\text{region}} + \beta_{\text{biome}} + \beta_{\text{Climate} \times \text{region}} + \beta_{\text{Climate} \times \text{biome}} \quad (1)$$

where y is moss cover, β_0 , β_{Climate} , β_{biome} , β_{region} , $\beta_{\text{Climate} \times \text{region}}$ and $\beta_{\text{Climate} \times \text{biome}}$ are the regression coefficients for the intercept, the fixed effects of climate, world region, biome and two-way interactions of climate with biome and region, respectively. We used the same effect sizes and weights assigned in the REM meta-analysis

(see the previous section), and we applied the restricted maximum-likelihood estimator (RMLE) method in the R package 'metafor' to fit REM meta-models (Viechtbauer, 2010). We avoided collinearity between predictors by examining their correlations and limiting any candidate model to only one predictor from any set of correlated predictors ($|r| \geq 0.5$) at a time. The most parsimonious model was selected using the Akaike information criteria for small sample sizes (AICc), which penalizes models with more terms, with equivalent models defined as those with the difference in their AICc values (ΔAICc) < 2 (relative to the best model; Anderson & Burnham, 2002).

For the US and AT datasets, the most parsimonious models were obtained by fitting beta-regression models (a special case of GLMs) for percentage values. Beta-regression models estimate the mean (μ) and its precision parameter (ϕ), each from a matrix of orthogonal predictors to describe the response variable (moss cover, y) as their function; $y = \text{Beta}(\mu, \phi)$, where the precision parameter adjusts the predicted values according to the dispersion of the modelled dataset (Cribari-Neto & Zeileis, 2010). Collinearity between the predictors was avoided, using the variance inflation factor (VIF), by iteratively removing the predictor with the highest VIF from the model and then recalculating VIF for the remaining predictors until all the VIFs were less than three (a threshold suggested for ecological studies by Zuur, Ieno, Walker, Saveliev, & Smith, 2009). The full models for the US (Equation 2) and AT (Equation 3) datasets were analysed with the R package 'betareg' (Cribari-Neto & Zeileis, 2010) using the RMLE method, with the most parsimonious models selected based on their AICc and equivalent models defined as those with $\Delta\text{AICc} < 2$ (Anderson & Burnham, 2002). The full model for the US dataset was defined as

$$f(y) = \beta_0 + \beta_{\text{Age}} + \beta_{\text{Canopy}} + \beta_{\text{Climate}} + \beta_* | \gamma_0 + \gamma_{\text{biome}} + \gamma_{\text{region}} \quad (2)$$

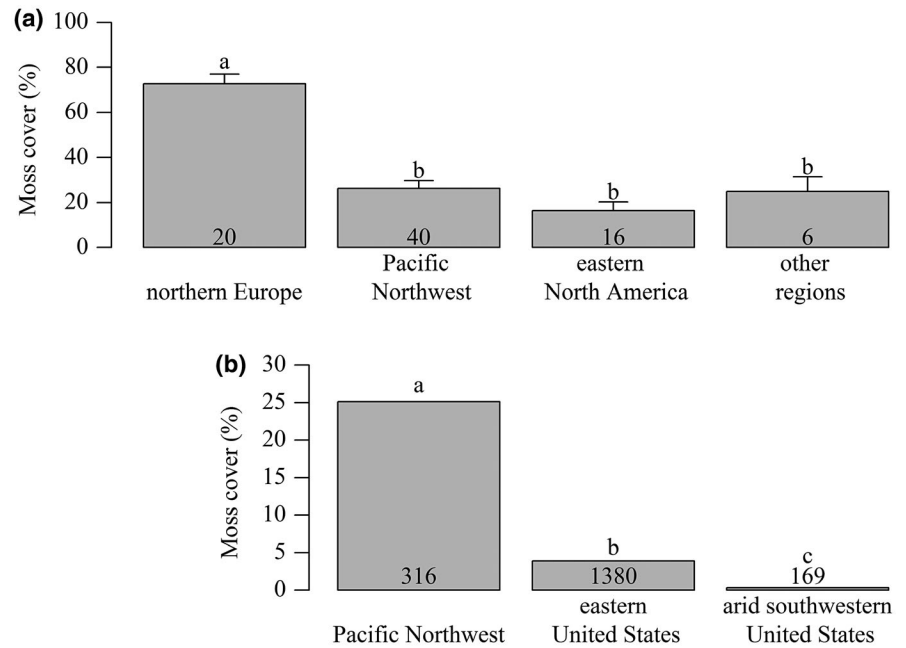
where y is moss cover, β_0 , β_{Age} , β_{Canopy} , β_{Climate} and β_* are the regression coefficients for the intercept, the fixed effects of forest age, canopy composition, climate and their two-way interactions, respectively and γ_0 , γ_{biome} and γ_{region} are the intercept, fixed effects of world region and biome, used to estimate the precision parameter. The full model for the AT dataset was defined as.

$$f(y) = \beta_0 + \beta_{\text{Canopy}} + \beta_{\text{Climate}} + \beta_{\text{Canopy} \times \text{Climate}} | \gamma_0 \quad (3)$$

where y is moss cover, β_0 , β_{Canopy} , β_{Climate} and $\beta_{\text{Canopy} \times \text{Climate}}$ are the regression coefficients for the intercept, main effects of canopy composition, climate and their interaction, and γ_0 is the default intercept to estimate the precision parameter.

Coefficients estimating the mean (β) were the focus of fitting beta-regression models for the US and AT datasets, whereas the coefficients estimating the precision parameter (γ) were considered only to account for dispersion. The predictor sets differed between full models based on the correlation threshold and on the availability of predictors in each dataset.

FIGURE 2 Moss cover in extratropical forests according to world region (bar labels) derived from the intercontinental-scale ETR dataset assembled using a global literature search (a) and from the separate continental-scale US dataset assembled using the systematic FIA survey in the United States (b). Error bars represent the mean $\pm$ SE; numbers inside or above the bars indicate n ; and different letters above the error bars indicate statistically different means. The length of the y axis varies between panels (a) and (b). Error bars in (b) were too narrow to show at the scale of the graph. Note that eastern North America in (a) is more broadly defined and based on a different set of plots than eastern United States in (b) (cf. Figure 1)



3 | RESULTS

Compared with the Pacific Northwest, eastern North America, eastern United States, arid southwestern United States and other extratropical regions of the world, northern Europe had the highest reported forest moss layer cover (Figures 1 and 2). The moss layer in northern Europe covered $72.7\% \pm 4.3\%$ (mean $\pm$ SE) of the forest floor, contrasting with $< 30\%$ in any other studied region (Figure 2a). Despite the differences in the number of observations and methodologies used in the ETR and US datasets (Table 1), these data sources agreed remarkably well that the moss layer covered c. 26% of forest floor in the only region that was common to these two datasets, namely the Pacific Northwest ($26.2\% \pm 3.6\%$ vs. $26.5\% \pm 1.5\%$, mean $\pm$ SE; Figure 2a,b). Moss cover on the forest floor in the Pacific Northwest was comparable to moss cover in eastern North America (including locations from southern Canada), but it was considerably greater than moss cover in eastern United States (Figure 2a,b). The arid southwestern United States had by far the least amount of moss (Figure 2b), as one could expect based on our hypothesis (hypothesis 1) that forest moss cover increases with latitude.

The ETR and US datasets spanned $>26^\circ$ of latitude and agreed that moss cover increased towards the North Pole (Figure 3a,b). Latitude accounted for between 40% (US dataset) and 46% (ETR dataset) of variation in forest moss cover (Supporting Information Appendix S3). Despite the differences in the character and sizes of the datasets (Table 1), both the literature search (ETR dataset) and the national systematic survey (US dataset) indicated an increment of c. 2% of forest moss cover per 1° of latitude in the Northern Hemisphere (Supporting Information Table S3.1 in Appendix S3). The systematic surveys (US and AT datasets) agreed on the means for moss cover in both conifer (*t*-test, $p = 0.13$) and broadleaf (*t*-test, $p = 0.96$) forests, although the literature search indicated a considerably higher moss cover in the broadleaf biome compared with the systematic surveys

(Figure 3c). Moss cover increased with latitude particularly within boreal and conifer forests, and less so in the temperate broadleaf biome, when analysed on broader scales (ETR and US datasets; Figure 4a–f), but moss cover did not respond strongly to latitude in the two biomes containing azonal forest vegetation (Figure 4g–h) or at more regional scales (AT dataset; Figure 4i).

Maximum temperature was the only climate variable consistently related to moss cover across all datasets; forest moss cover in all datasets decreased with increasing maximum temperature (all p -values < 0.001 ; Figure 5; Supporting Information Table S3.3 in Appendix S3). Maximum temperature accounted for 13% to 36% of variation in forest moss cover (depending on the dataset), with moss cover declining by 2% to 5% per 1°C increase in maximum temperature (Figure 5; Supporting Information Table S3.4 in Appendix S3). Although multivariate analyses confirmed that climate (maximum temperature) was an important driver of moss cover at all three spatial scales (ETR, US, and AT), the effect of maximum temperature on moss cover was one order of magnitude smaller than the effects of predictors such as region, biome and forest stand composition (conifer vs. broadleaf; Figure 6; Supporting Information Tables S3.5 and S3.6 in Appendix S3). Thus, the moss cover changes documented over the latitudinal climatic gradient (Figures 3 and 5) appear to reflect, at least in part, changes in environmental conditions and forest composition (from broadleaf to conifer) between the regions and biomes arranged along this latitudinal climatic gradient.

4 | DISCUSSION

The effects of latitudinal gradients in climate on the abundance of forest moss in our study are consistent with latitudinal abundance patterns caused by gradients in temperature, moisture and human impact observed for some other plant taxa (e.g., trees; Crowther et al., 2015). The higher forest moss abundance in northern Europe

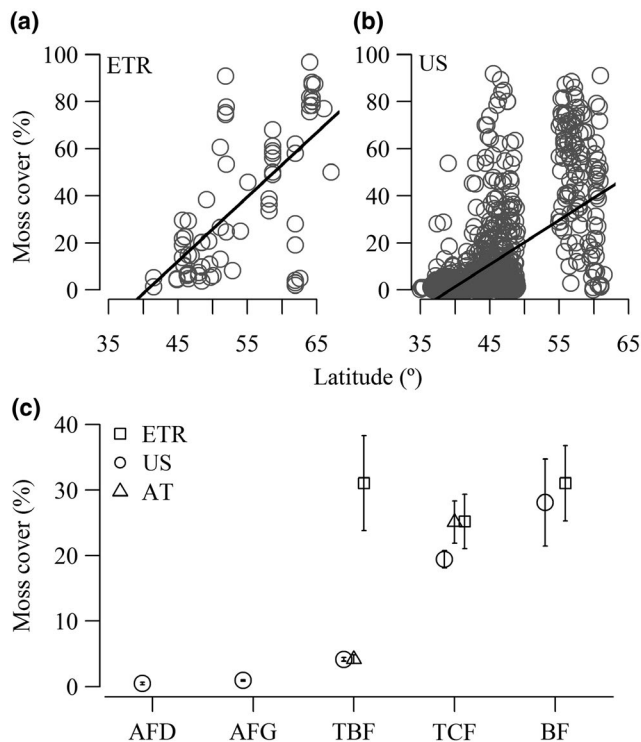


FIGURE 3 Forest moss cover along the latitudinal span of the intercontinental-scale ETR dataset assembled using a literature search (a), along the latitudinal span of the continental-scale US dataset assembled using the systematic FIA survey in the United States (b), and across the analysed biomes (c) of each individual dataset (ETR, US, and AT). Note the data gap in (b) between c. 50 and 55° N attributable to a discontinuity in the FIA plot locations with moss layer cover recorded. Error bars represent the mean $\pm$ SE. In (c), biomes were based on Olson et al. (2001), and community abbreviations are as follows: AFD = azonal forests in the desert and xeric shrublands biome; AFG = azonal forests in the temperate grassland, savannas and shrublands biome; BF = boreal forest; TBF = temperate broadleaf forest; TCF = temperate conifer forest

(ETR dataset) is consistent with the latitudinal gradient in forest moss abundance that we documented based on the systematic survey at the continental scale (US dataset). This latitudinal pattern in forest moss cover is broadly similar to the pattern of moss species richness described for other extratropical areas (cf. Europe; Mateo et al., 2016). In contrast, the patterns of increasing moss abundance and richness with latitude are not fully consistent with latitudinal patterns in moss species density across 28 Chinese provinces spanning both tropical and extratropical biomes (Chen et al., 2015), probably because the positive relationship between moss richness and latitude in temperate (extratropical) regions can be veiled at larger scales by mean abundance and turnover of moss flora in the tropics (cf. Hedenäs, 2007). Our findings suggest that latitude and latitudinal span matter when searching for moss abundance patterns at broad spatial scales, as also observed for latitudinal patterns of moss richness (Shaw, Cox, & Goffinet, 2005).

The consistent negative correlation between moss cover and maximum temperature that we observed seems somewhat contradictory to

the notion that moss cover is sensitive mainly to moisture availability and less so to temperature at broad spatial scales (e.g., Mateo et al., 2016). However, maximum temperature drives evapotranspiration, and thus it can affect ecosystem water balance and lead to moss desiccation (Nichols & Brown, 1980). Given that the effect of moisture availability on moss cover has been documented for ground-level mosses, which respond to both vapour pressure deficit and soil moisture (e.g., Stewart & Mallik, 2006), maximum temperature can be used as a proxy for moisture limitation in studies of mosses in extratropical forests at broad spatial scales (cf. Sandholt, Rasmussen, & Andersen, 2002). Indeed, temperature was shown to have a negative influence on species richness across Canadian forests, where latitudinal climatic gradients explained 19% of the variability, although lichens were pooled with mosses in that study (Zhang, Chen, & Taylor, 2014), potentially obscuring moss distribution patterns (cf. Cornelissen et al., 2007). Our analyses might also suggest that precipitation might not be the best proxy for moisture availability for forest mosses (but see Belnap, Phillips, & Troxler, 2006); instead, field measurements or more complex indices that include relative humidity, vapour pressure deficit, effects of local topography, soil characteristics, or potential evapotranspiration might be more useful (cf. Karger et al., 2012).

At the continental and subcontinental scales, forest moss cover appeared to be driven to a large degree by canopy composition (broadleaf and conifer trees), corroborating conclusions from studies of moss cover on finer spatial scales (e.g., Cavard, Bergeron, Chen, & Pare, 2011). Importantly, our analyses suggested that forest moss cover reported in the literature for temperate broadleaf forests tends to overestimate moss cover compared with the estimates obtained by systematic surveys, suggesting a potential bias towards higher moss cover values in studies focused on moss communities of temperate broadleaf forests—possibly attributable to selection of moss-rich sites or microsites in areas where moss cover is variable but generally less abundant (Cavard et al., 2011). Then, the spatial heterogeneity of the moss layer might be higher compared with that of its hosting forest, as expected from the variability of spatial heterogeneity across scales (Gigante, Foggi, Venanzoni, Viciani, & Buffa, 2016; Palmer, 1988). Notably, the bias of published studies towards higher moss abundance was not present in the boreal coniferous forests, where moss was much more abundant than in the temperate broadleaf forests (cf. Bailey, 2014).

The fact that maximum temperature modulated moss cover across latitudinal gradients at all three broad scales in our study (intercontinental, continental, and subcontinental) suggests that moss cover is a promising tool for monitoring moss community responses to global warming across diverse ecoregions and biomes (cf. He et al., 2016; Mateo et al., 2016). Several features of forest moss layer species—their small body size, shorter life span and sensitivity to heat (moisture) stress; Glime, 2007—suggest that the moss layer might potentially have a faster response to climate warming compared with longer lived taxa, such as trees. Importantly, monitoring moss cover changes forced by climate warming or by other environmental factors allows insight into changes in ecosystem-level processes, such as water and carbon storage or nutrient budgets, in which mosses play important roles (Collins et al., 2013). Monitoring the moss layer might provide early information on several interrelated aspects of ongoing changes in forest ecosystems. For example, predicted climate warming (see <https://nca2014.globalchange.gov/>, Olsen et al., 2013;

FIGURE 4 Changes in forest moss cover with latitude within the studied biomes (as defined by Olson et al., 2001): boreal forest (BF in a and d), temperate conifer forest (TCF in b, e and i), temperate broadleaf and mixed forests (TBF in c, f and i), azonal forests in temperate grasslands and savanna biome (AFG in g) and azonal forests within desert and shrubland biome (AFD in h). The individual graphs were derived from the intercontinental-scale ETR dataset assembled using a literature search (a–c), from the continental-scale US dataset assembled using the systematic FIA survey in the United States (d–h) and from the subcontinental-scale AT dataset assembled from a systematic survey along the Appalachian Trail in eastern United States (i). Note that latitudinal ranges on the x axis vary depending on the plot locations. A summary of the corresponding simple random effect model (REM) meta-regressions (a–c) and of the simple linear regressions (d–i) is shown in Supporting Information Table 3.2 in Appendix S3

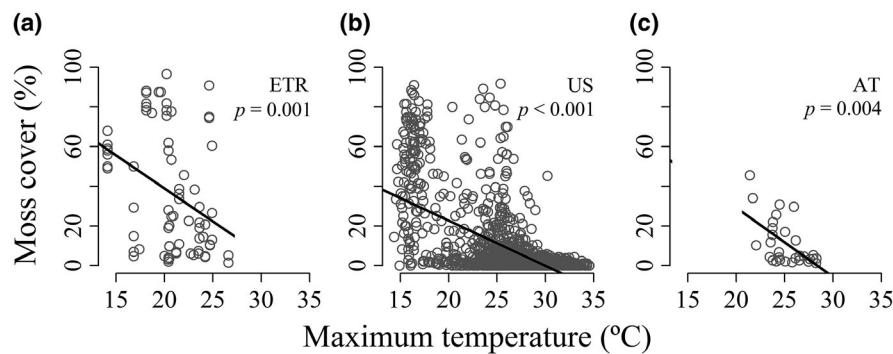
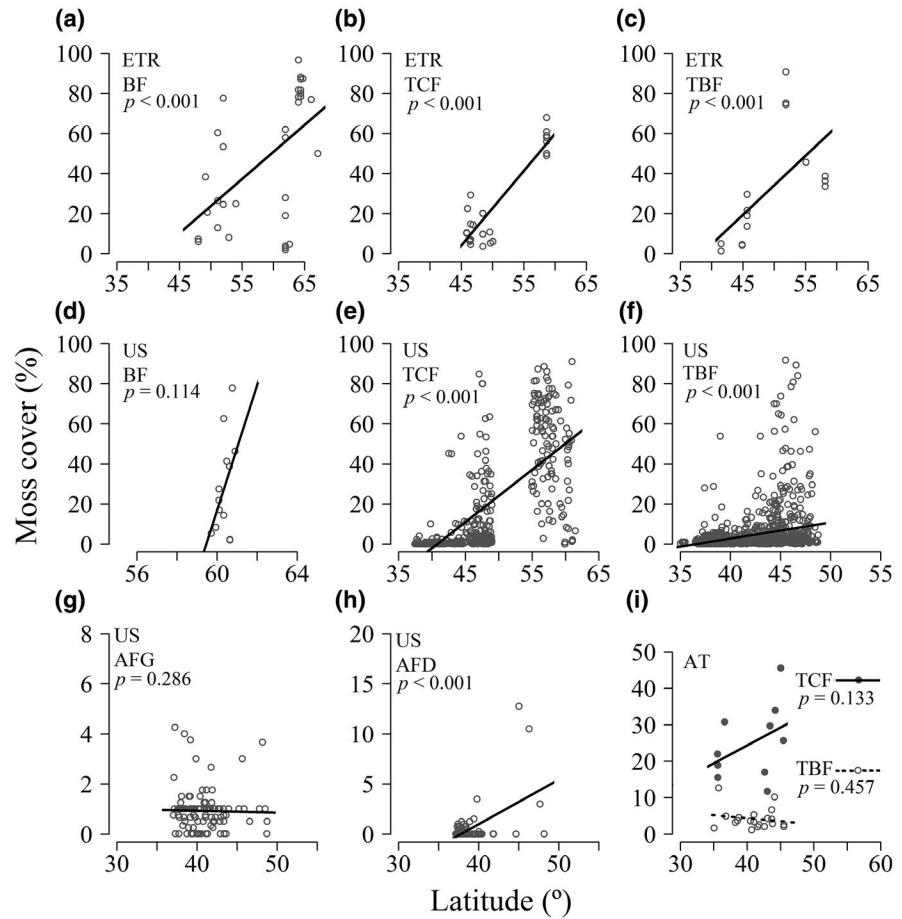


FIGURE 5 Changes in forest moss cover with maximum temperature derived from the intercontinental-scale ETR dataset assembled using a literature search (a), the continental-scale US dataset assembled using the systematic FIA survey in the United States (b) and the region-wide subcontinental-scale systematic AT dataset from the eastern United States (c). A summary of the corresponding simple random effect model (REM) meta-regressions (a) and of the simple linear regressions (b and c) is shown in Supporting Information Appendix S3

Brown & Caldeira, 2017) may cause decline in moss cover, which can be magnified further by land management- or climate-induced changes in forest canopy composition (Lambrechts, Wilkie, Rucevska, & Sen, 2009). In turn, the decline in moss cover (He et al., 2016) may increase carbon emissions from forested ecosystems (Deane-Coe & Stanton, 2017) and contribute to further climate warming (cf. DeLucia et al., 2003).

In conclusion, our findings allow us to suggest that measurements of moss cover can contribute to monitoring forest vegetation change over time, and particularly so under global climate change. Forest moss

cover measurements are quick and have often been implemented as a component of vegetation inventories carried out at forest stand scales in research studies and national or regional monitoring programmes. Importantly, our results allow us to draw three important recommendations. First, there is a clear need to elucidate whether the latitudinal patterns and drivers of the moss layer documented in North America and Europe also occur in relatively less studied South America and Asia (see also Cornelissen et al., 2007; Geffert, Frahm, Barthlott, & Mutke, 2013; Hedenäs, 2007). Second, data from specialized literature should

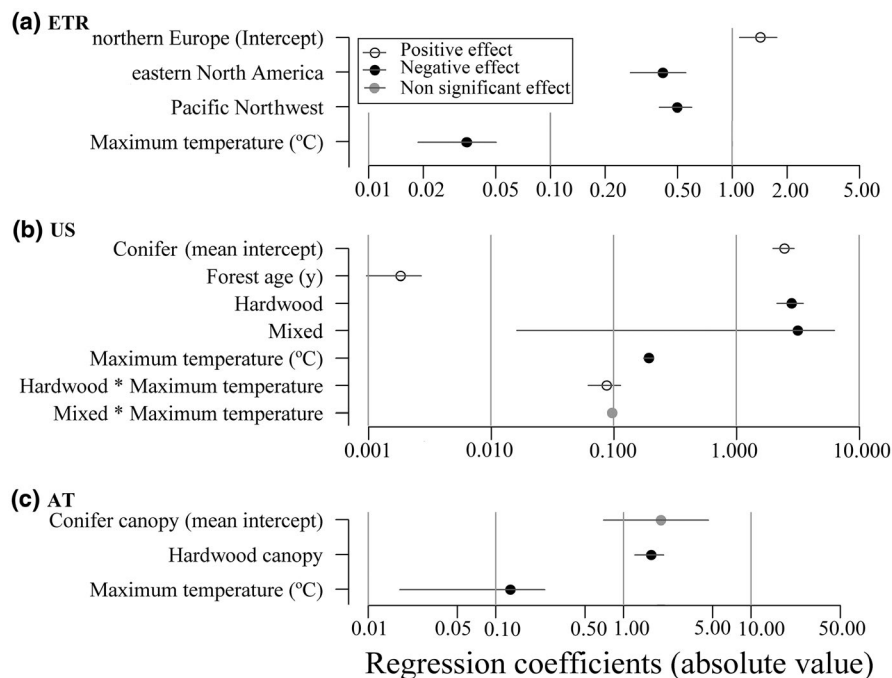


FIGURE 6 Coefficients of the most parsimonious models fitted to the intercontinental-scale ETR dataset assembled using a literature search (a), the continental-scale US dataset assembled using the systematic FIA survey (b) and the region-wide subcontinental-scale systematic AT dataset from the eastern United States (c) to test the effects of major environmental drivers (forest age, canopy composition, and climate; see Table 2) when accounting for the effects of biome and world region on forest moss cover at broad scales. Note that x axes are on logarithmic scales and depict absolute values of the predictor coefficients; open circles indicate positive effects on moss cover, filled circles indicate negative effects on moss cover, and grey circles indicate non-significant effects. Error bars show estimated 95% confidence intervals. Beta regressions (b and c) include only the regression coefficient estimating the mean. Supporting Information, Tables S3.5 and S3.6 in Appendix S3, summarizes the corresponding random effect model (REM) meta-regressions (a) and beta-regressions (b and c)

be complemented by systematic surveys in ecosystems with low and variable abundance of the taxa of interest (e.g., moss layer in temperate broadleaf forests). Third, publishing data on moss cover (in public repositories, data banks, or scientific papers) can consolidate the potential of moss cover as a monitoring tool and can contribute to our knowledge of moss layer patterns at broad spatial scales.

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DATA ACCESSIBILITY

The three datasets with the compiled forest moss cover data are available as Supporting Information (Appendix S1).

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

BIOSKETCH

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